MAgent Theory 2.0

Supplementary Proofs

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This supplement gives supporting calculations and secondary proofs. Each result restates its assumptions and uses the same notation as the main text. The main constructions and their decisive proofs remain in the main manuscript and its appendices.

Chapter S1

Probability and Predictive Reconstruction

S1.1 The conditional relative-entropy identity

The disintegrations used here are regular conditional laws on standard Borel spaces [7].

Let μ, ν be probability laws on a product $X \times Y$ of standard Borel spaces. Write their marginals as μ_Y, ν_Y and their conditional laws as μ^y, ν^y. Then

$$D_{\mathrm{KL}}(\mu\|\nu) = D_{\mathrm{KL}}(\mu_Y\|\nu_Y) + \int D_{\mathrm{KL}}(\mu^y\|\nu^y)\mu_Y(dy). \tag{S1.1}$$

The identity holds in $[0,\infty]$, both sides being infinite together. Suppose first $\mu \ll \nu$ and put $f = d\mu/d\nu$, a jointly measurable function. Then $a(y) = \int f(x,y)\nu^y(dx)$ is a version of $d\mu_Y/d\nu_Y$, and $b = f/a$ on $\{a > 0\}$ is jointly measurable with $\mu^y = b(\cdot, y)\nu^y$ for μ_Y-almost every y. Starting from f in this way avoids choosing a jointly measurable version of the conditional densities separately. The logarithm splits into $\log a + \log b$. The negative parts of $x \log x$ are bounded by $1/e$, so the negative parts of the two logarithms have finite μ-integrals and the split integrates in $[0,\infty]$, giving eq. (S1.1). If μ is not absolutely continuous with respect to ν, then either $\mu_Y \not\ll \nu_Y$ or $\mu^y \not\ll \nu^y$ on a set of positive μ_Y-measure, and both sides are infinite. This argument underlies the predictive and reconstruction identities in the main text.

S1.2 Different predictive mechanisms

Let s, t be laws on a model space and let K, L be normalized kernels into a common record space. Suppose the following joint divergence is finite. Define

$$\mu(dm, do) = s(dm)K(m, do), \qquad \nu(dm, do) = t(dm)L(m, do). \tag{S1.2}$$

Disintegration first by m and then by o gives the exact equality

$$\begin{aligned} D_{\mathrm{KL}}(s\|t) + \int D_{\mathrm{KL}}(K(m,\cdot)\|L(m,\cdot))s(dm) &= D_{\mathrm{KL}}(sK\|tL) \\ &\quad + \int D_{\mathrm{KL}}(\mu^o\|\nu^o)(sK)(do). \end{aligned} \tag{S1.3}$$

This is two uses of eq. (S1.1) on one joint law. It displays explicitly the distinction contributed by the evaluators. Equal model laws do not remove that term.

For a bijective change of model presentation Θ, replace t by $\Theta_{\#}t$ and replace L by $L(\Theta^{-1}m, \cdot)$. Both sides then compare laws in one presentation. Measurable record relabeling must be applied consistently as well.

S1.3 A conditional reconstruction on four points

Let $E = \{0,1\}^2$, write its coordinates as (H, V), and transmit only H. Choose a sender law with

$$q(H = 1) = \frac{1}{2}, \qquad q(V = H \mid H) = 1. \tag{S1.4}$$

Let the receiver's prior p make H, V independent uniform bits. The communicated marginal is uniform, so reconstruction using $p(V \mid H)$ returns the uniform law on all four points. Its transmitted marginal is exactly correct, but it discards the sender's perfect dependence. Direct evaluation gives

$$D_{\mathrm{KL}}(q\|\widehat{q}) = 2\left(\frac{1}{2}\log\frac{1/2}{1/4}\right) = \log 2. \tag{S1.5}$$

The conditional divergence at each value of H is also $\log 2$, verifying the conditional identity exactly. A receiver whose prior has the same conditional relation $V = H$ instead reconstructs q from the transmitted marginal. The difference comes from its prior conditional, not from a larger message. The reconstruction is Jeffrey's rule of conditioning the receiver's prior on the transmitted marginal [4], and it is the law closest to that prior, in relative entropy, among all laws with the transmitted marginal.

S1.4 A shared source and two routes

Let X be a uniform bit. Two routes return $Y_1 = X$ and $Y_2 = X \oplus b$, where $b \in \{0,1\}$ is a fixed route mark and $\oplus$ is addition modulo two. The comparison $O = Y_1 \oplus Y_2$ equals b with probability one. If the second route instead uses an independent uniform bit X', the same comparison is uniform and reveals no route mark. The marginal law arriving on each route is uniform in both experiments. Thus the shared-source dependence, not those marginals, determines the comparison outcome. A record model intended to inspect a loop must specify which occurrences refer to the same source. In general, a functional of the pair of route laws cannot distinguish two loop maps H, H' with the same pushforward of the source law, whereas the joint law of a shared sample and its image along the loop determines the loop map almost everywhere; the route mark is the simplest instance.

S1.5 Predictive sufficiency for a selected response

Let $X = (A, B)$ be two independent uniform bits, let the encoder retain $U = A$, and take no side information. For target $Y = A$, one has $I(Y; X \mid U) = 0$. For target $Y = B$, the same representation gives $I(Y; X \mid U) = \log 2$. Perfect predictive retention is therefore a statement about a specified target. If a later interaction reads B, the first sufficiency statement does not supply its response. This elementary distinction persists when X is a complete configuration of law-valued agents and Y is a boundary process.

Chapter S2

Algebra of Exponential Links with Low-Rank Generators

This chapter gives supporting matrix identities for interactions written in a reference vector space. All spaces are finite dimensional and real. The results concern arbitrary finite matrices and do not require Gaussian laws, diagonalizability, or invertibility of the generator. A statistical interpretation enters only when the matrices are assigned to interfaces or made functions of law parameters.

S2.1 The low-rank exponential

Let $O : \mathbb{R}^r \to \mathbb{R}^d$ be a receiving map and $T : \mathbb{R}^d \to \mathbb{R}^r$ a sending map. Define the entire matrix function

$$\varphi_1(Z) = \sum_{n=1}^{\infty} \frac{Z^{n-1}}{n!}. \tag{S2.1}$$

When Z is invertible this equals $(e^Z - I)Z^{-1}$, but the series defines it also at singular Z; it is the first of the φ functions of exponential integrators [6, Chapter 10].

Lemma S2.1 (Exponential through the message space). The following identity holds for all O, T:

$$e^{OT} = I_d + O\varphi_1(TO)T. \tag{S2.2}$$

Proof. Associativity gives $(OT)^n = O(TO)^{n-1}T$ for every integer $n \geq 1$. Absolute convergence of the finite-dimensional exponential series permits summation and yields eq. (S2.2). □

This is the classical relation between $f(AB)$ and $f(BA)$ for $f = \exp$, in the form $f(AB) = f(0)I + Ag(BA)B$ with $g(z) = (f(z) - f(0))/z$, which needs no nonsingularity hypothesis [6, Section 1.8]. In particular, the correction to identity has image in $\operatorname{im} O$ and vanishes on $\ker T$. The image of the full exponential need not be low dimensional. Indeed its inverse is e^{-OT}, so the identity contribution preserves all input dimensions even when the communicated correction is selective.

The identity also applies to several interfaces simultaneously. For maps O_h, T_h, stack $O = [O_1 \ \cdots \ O_H]$ and stack the T_h vertically to form T. Then $OT = \sum_h O_h T_h$ and the same formula holds. The smaller matrix TO includes all cross-interface products $T_h O_k$; replacing it by its diagonal blocks would require additional vanishing conditions.

Proposition S2.2 (Rank-one closed form and determinant). For column vectors $u, v \in \mathbb{R}^d$, put $L = uv^T$ and $t = v^T u$. Then

$$e^L = I_d + \varphi_1(t)L, \qquad \det e^L = e^t. \tag{S2.3}$$

If $t = 0$, then $L^2 = 0$ and $e^L = I_d + L$.

Proof. The first identity is eq. (S2.2) with a one-dimensional message space. For any square matrix A, Jacobi's determinant identity applied to $X(s) = e^{sA}$ gives $\frac{d}{ds} \det X(s) = \operatorname{tr}(A) \det X(s)$. Since $X(0) = I$, its determinant is $e^{s \operatorname{tr}(A)}$. Set $A = L$, $s = 1$, and use $\operatorname{tr}(uv^T) = v^T u$. Finally $L^2 = u(v^T u)v^T$ proves the nilpotent case. □

This result distinguishes a singular generator from a singular interaction: a rank-one generator gives an invertible exponential in every dimension. It does not assert that every invertible real matrix is one exponential: a real matrix is a real exponential exactly when it is invertible and, for every negative eigenvalue λ and every size k, the Jordan blocks $J_k(\lambda)$ occur an even number of times [3]. Counting all blocks of a negative eigenvalue together gives wrong answers: $J_1(-1) \oplus J_3(-1)$ and $\operatorname{diag}(-1, -2)$ have no real logarithm.

S2.2 Differentiating a law-dependent link

Lemma S2.3 (Derivative of the matrix exponential). For square matrices L, H, the Fréchet derivative is

$$D \exp_L[H] = \int_0^1 e^{(1-t)L} H e^{tL} dt. \tag{S2.4}$$

Proof. Let $X_\epsilon(t) = e^{t(L+\epsilon H)}$. Differentiating its matrix differential equation with respect to ϵ at zero gives $\dot{Y} = LY + He^{tL}$, $Y(0) = 0$. Multiplying by e^{-tL} and integrating gives $Y(t) = \int_0^t e^{(t-s)L} H e^{sL} ds$. Evaluation at $t = 1$ proves the formula. Smooth dependence is also obtained directly by differentiating the locally uniformly convergent exponential series. The integral representation is the Duhamel formula of Wilcox [8]; see also Higham [6, Chapter 10]. □

If $[L, H] = 0$, this derivative reduces to $e^L H$. Without that hypothesis the replacement is invalid in general. For a link $\Theta = U_i e^L U_j^{-1}$ with fixed endpoint frames, its variation is $U_i D \exp_L[dL] U_j^{-1}$. If the frames vary, the product rule supplies their two additional terms, including $d(U_j^{-1}) = -U_j^{-1}(dU_j)U_j^{-1}$.

For the moment-dependent family $L = \lambda \mu_i \mu_j^T \Sigma_j^{-1}$, with fixed scalar λ, ordinary differentiation gives

$$\begin{aligned} dL = \lambda \big[&(d\mu_i)\mu_j^T \Sigma_j^{-1} + \mu_i (d\mu_j)^T \Sigma_j^{-1} \\ &- \mu_i \mu_j^T \Sigma_j^{-1} (d\Sigma_j) \Sigma_j^{-1} \big]. \end{aligned} \tag{S2.5}$$

Here a differentiable family of laws is assumed to have differentiable first and second moments; differentiation under their defining integrals requires the corresponding domination or integrability hypotheses. Positive-definite covariance makes inversion smooth. Combining eqs. (S2.4) and (S2.5) gives the complete link derivative on this admitted family, with no assumption about the shape of its densities. That statement concerns $d\Theta$ only. Differentiating a cost such as $D_{\mathrm{KL}}(\nu_i \| \Theta_\# \nu_j)$ in Θ at Θ_0 needs hypotheses on the shapes of the laws. Sufficient conditions are that $\nu_i \ll \Theta_\# \nu_j$ with finite divergence for Θ near Θ_0, that ν_j has a continuously differentiable positive density p_j on the relevant region, and that $\mathbb{E}_{\nu_i} \sup_\Theta \big| \partial_\Theta \big[\log p_j(\Theta^{-1} Y) - \log |\det \Theta| \big] \big| < \infty$ over a neighborhood of Θ_0; dominated differentiation then applies to the only Θ-dependent term $-\mathbb{E}_{\nu_i} \log \big(p_j(\Theta^{-1} Y) / |\det \Theta| \big)$.

Differentiability in quadratic mean of the family $\Theta_{\#}\nu_j$, which for linear Θ is the scale-and-shear orbit of ν_j, is the condition for the second-order expansion at the diagonal; it is not sufficient for first differentiability of the cost, since a two-point ν_i is singular to every $\Theta_{\#}\nu_j$ of a Gaussian ν_j, and the sufficient conditions above do not require it. Without such hypotheses differentiability can fail within the admitted moment family: if ν_j is uniform on $[1,2]\times[0,1]$ and $\nu_i = (\Theta_0)_{\#}\nu_j$, then for $\Theta = (1\pm\epsilon)\Theta_0$ the support of ν_i is not contained in that of $\Theta_{\#}\nu_j$, so the divergence is infinite arbitrarily close to Θ_0, where it vanishes.

Under such hypotheses, the resulting terms enter a variational derivative whenever an interaction energy reads the link. Holding the generator fixed suppresses them and defines a different update convention. This algebra separates the constitutive choice of a law-dependent interaction from the kinetic choice used to evolve the laws; neither choice follows from the exponential parameterization alone.

Chapter S3

Gaussian comparison costs and repeated variables

This chapter distinguishes a Gaussian comparison likelihood from a divergence between Gaussian laws by direct calculation. The comparison likelihood is normalized in its record. Its expectation depends on the joint law of the contents that enter the comparison, including whether two appearances denote the same variable. The calculations require finite second moments; Gaussian recognition is needed only where a Gaussian relative entropy is evaluated.

S3.1 The expected comparison likelihood

Let $X_i \in \mathbb{R}^{n_i}$ and $X_j \in \mathbb{R}^{n_j}$ have means μ_i, μ_j, covariances Σ_{ii}, Σ_{jj}, and cross covariance $\Sigma_{ij} = \operatorname{Cov}(X_i, X_j)$. Let $A : \mathbb{R}^{n_i} \to \mathbb{R}^d$ and $B : \mathbb{R}^{n_j} \to \mathbb{R}^d$ be fixed linear maps. For $R \succ 0$, define the normalized record law

$$O \mid X_i, X_j \sim \mathcal{N}(AX_i - BX_j, R), \qquad c_R = \tfrac{1}{2} \log\det(2\pi R). \tag{S3.1}$$

Proposition S3.1 (Complete residual covariance). At a fixed record o, set $\delta = A\mu_i - B\mu_j - o$. Then

$$\begin{aligned} \mathbb{E}[-\log p(o \mid X_i, X_j)] &= c_R + \tfrac{1}{2}\delta^T R^{-1}\delta + \tfrac{1}{2}\operatorname{tr}(R^{-1}V_{ij}), \\ V_{ij} &= A\Sigma_{ii}A^T + B\Sigma_{jj}B^T - A\Sigma_{ij}B^T - B\Sigma_{ji}A^T. \end{aligned} \tag{S3.2}$$

Proof. The negative log density is $c_R + \frac{1}{2}(AX_i - BX_j - o)^T R^{-1}(AX_i - BX_j - o)$. Write its residual as its mean δ plus a centered variable. The linear cross terms have zero expectation, and the centered quadratic has expectation $\operatorname{tr}(R^{-1}V_{ij})$. Expanding the covariance of $AX_i - BX_j$ gives the stated four terms. □

Independence is sufficient to remove Σ_{ij}, but it is stronger than necessary. Only the cross covariance visible through A and B contributes to this cost. A pair can be dependent while both displayed cross terms vanish. Conversely, knowing the two marginal laws does not generally determine the expected likelihood.

S3.2 Equal rows need not give equal covariance updates

For the next calculation let the receiver law p of AX_i in the comparison space have mean a, covariance $V \succ 0$, and finite differential entropy $H(p)$, let $BX_j \sim \mathcal{N}(b_j, W_j)$ with $W_j \succ 0$, assume $A\Sigma_{ij}B^T = 0$ (independence is sufficient), and take $o = 0$. By proposition S3.1, which needs only

second moments, the likelihood score is

$$C_j = c_R + \tfrac{1}{2}\left[(a-b_j)^T R^{-1}(a-b_j) + \mathrm{tr}(R^{-1}(V+W_j))\right]. \tag{S3.3}$$

The direct peer score is $D_j = D_{\mathrm{KL}}(p\|\mathcal{N}(b_j, W_j)) = -H(p) + c_{W_j} + \frac{1}{2}[(a-b_j)^T W_j^{-1}(a-b_j) + \mathrm{tr}(W_j^{-1}V)]$. For a Gaussian receiver $p = \mathcal{N}(a, V)$ it reads

$$D_j = \tfrac{1}{2}\left[(a-b_j)^T W_j^{-1}(a-b_j) + \mathrm{tr}(W_j^{-1}V) - d + \log\frac{\det W_j}{\det V}\right]. \tag{S3.4}$$

This is the divergence $D_{\mathrm{KL}}(A_\# q_i \| B_\# q_j)$ between the two laws in the common comparison space; it equals the peer divergence of the dynamics, the divergence from the receiver law to the channelled source law, only when A is invertible and the channel is identified accordingly. Its receiver quadratic has expectation d, while its source quadratic has expectation $\mathrm{tr}(W_j^{-1}V) + (a-b_j)^T W_j^{-1}(a-b_j)$.

Proposition S3.2 (A restricted routing equivalence). Suppose $W_j = W$ for every source, W is held fixed, and $R = TW$ for $T > 0$. Then

$$C_j - \frac{D_j}{T} = c_{TW} + \frac{d}{2T} + \frac{H(p) - c_W}{T}, \tag{S3.5}$$

which does not depend on j. Consequently normalized rows proportional to $\pi_j e^{-C_j}$ and to $\pi_j e^{-D_j/T}$ coincide. For a Gaussian receiver, $H(p) = c_V + d/2$ and

$$C_j - \frac{D_j}{T} = c_{TW} + \frac{d}{T} - \frac{1}{2T}\log\frac{\det W}{\det V}. \tag{S3.6}$$

For every affine receiver family $AX_i = a + M\varepsilon$, with a fixed standardized ε of finite entropy and $V = MM^T$, the derivatives of the two scores with respect to receiver covariance nevertheless differ by

$$d_V C_j - d_V(D_j/T) = \frac{1}{2T}\,\mathrm{tr}(V^{-1}dV). \tag{S3.7}$$

Proof. Insert $R = TW$ and $W_j = W$. The mean quadratic and $\mathrm{tr}(W^{-1}V)$ cancel in $C_j - D_j/T$, leaving eq. (S3.5), which is independent of j. A common additive score cancels on row normalization. For Gaussian p, substitution of $H(p) = \frac{1}{2}\log\det(2\pi e V)$ gives eq. (S3.6). In the affine family $H(p) = H(\varepsilon) + \frac{1}{2}\log\det V$, and differentiating the remaining $\log\det V$ gives eq. (S3.7). □

The zero record is needed: for $o \neq 0$ the difference acquires the term $-(a-b_j)^T W^{-1} o/T$, which depends on j. At $T = 1$ the Gaussian-receiver score difference is $H(\mathcal{N}(a, V)) + d/2$, with entropy measured against Lebesgue measure. It is constant for source selection, but not for variation of the receiver law. The receiver-mean and source-mean derivatives agree for every V, since eq. (S3.5) contains neither a nor b_j; only the receiver-covariance derivatives differ. If source covariances vary independently, or the comparison maps change them, the hypothesis of a common held W no longer applies. These are restrictions on the optimization block, not alternative formulas for one universal interaction.

S3.3 Identity self-comparisons and independent snapshots

Take $A = B = I_d$ and let a self comparison use the same random variable twice. Then $X_i - X_i = 0$ identically, so

$$\mathbb{E}[-\log p(o \mid X_i, X_i)] = c_R + \tfrac{1}{2} o^T R^{-1} o. \tag{S3.8}$$

Its derivative with respect to the law of X_i is zero. In eq. (S3.2), all four covariance terms cancel because Σ_{ii} is also the cross covariance. The zero-record identity peer divergence $D_{\mathrm{KL}}(q_i \| q_i) = 0$ likewise has zero total derivative.

Now let X_i and $\widetilde{X}_i$ be independent variables with the same $\mathcal{N}(0, I_d)$ law, and compare their difference with noise $R = TI_d$. At zero record their cost is $c_R + d/T$. Thus an identity self candidate and a distinct independent candidate with the same marginal law receive different likelihood scores, while their peer divergences are both zero. With equal source priors, the likelihood odds in favor of the identity self candidate are $e^{d/T}$.

An independently sampled self snapshot realizes the second calculation. A remembered state can instead be correlated with the current state, in which case its joint law determines an intermediate or different cost through eq. (S3.2). Calling a candidate "self" does not select that joint law. The distinction must be fixed before computing a comparison or a receiving update.

S3.4 Integrating the source before taking a logarithm

Let $X \sim \mathcal{N}(a, V)$, $Y \sim \mathcal{N}(b, W)$ be independent, with $V \succ 0$, $W \succeq 0$, and $R \succ 0$. Regard $k(x \mid y) = \mathcal{N}(x; y, R)$ as a normalized receiver-coordinate kernel. Integrating over the sender gives $r(x) = \mathcal{N}(x; b, W + R)$. Therefore

$$-\mathbb{E}_X \log r(X) = c_{W+R} + \tfrac{1}{2}\left[(a-b)^T (W+R)^{-1}(a-b) + \operatorname{tr}((W+R)^{-1}V)\right]. \tag{S3.9}$$

This expression compares the receiver to a noise-convolved source. It is not the peer divergence against $\mathcal{N}(b, W)$, even when that law is nonsingular.

Subtracting eq. (S3.9) from the expected negative log kernel gives the exact Jensen gap

$$\begin{aligned} G = {} & \tfrac{1}{2} \log \frac{\det R}{\det(W+R)} + \tfrac{1}{2} \operatorname{tr}(R^{-1}W) \\ & + \tfrac{1}{2}(a-b)^T \left[R^{-1} - (W+R)^{-1}\right](a-b) \\ & + \tfrac{1}{2} \operatorname{tr}\left(\left[R^{-1} - (W+R)^{-1}\right] V\right). \end{aligned} \tag{S3.10}$$

For completeness, $R^{-1} - (W+R)^{-1}$ is positive semidefinite: conjugating by $R^{1/2}$ reduces this assertion to $I - (I + R^{-1/2} W R^{-1/2})^{-1} \succeq 0$. The first line of eq. (S3.10) is half the sum of $\lambda - \log(1+\lambda)$ over the nonnegative eigenvalues of $R^{-1/2} W R^{-1/2}$. Each summand is nonnegative because its derivative is $\lambda/(1+\lambda)$ and its value at zero is zero. Hence $G \geq 0$, with $G > 0$ whenever $W \neq 0$. Averaging the conditional log density and taking the log of the integrated density remain different even at coincident means.

Chapter S4

Supporting Constructions for Inheritance

This chapter gives two exact ways to determine what a parent must retain: factorization of a boundary map and conditional prediction from a retained history. Both distinguish an inherited response from a newly chosen parent interaction.

S4.1 Linear boundary descent

Proposition S4.1 (Descent through a surjective retention). Let E_j, E_i, Z_j, Z_i be finite-dimensional real vector spaces. Let $C_j : E_j \to Z_j$ be surjective, let $C_i : E_i \to Z_i$ be linear, and let $M_{ij} : E_j \to E_i$ be a linear boundary map. There is a linear map $M^c_{IJ} : Z_j \to Z_i$ satisfying

$$C_i \circ M_{ij} = M^c_{IJ} \circ C_j, \tag{S4.1}$$

if and only if

$$\ker C_j \subseteq \ker(C_i \circ M_{ij}). \tag{S4.2}$$

When it exists, it is unique.

Proof. If eq. (S4.1) holds, every $v \in \ker C_j$ satisfies $C_i(M_{ij}v) = 0$. Conversely, define $M^c_{IJ}(C_j v) = C_i(M_{ij}v)$. Two choices v, w with $C_j v = C_j w$ differ by an element of $\ker C_j$, which eq. (S4.2) annihilates. The definition is therefore independent of the representative. It is linear, and surjectivity of C_j defines it everywhere. The same surjectivity proves uniqueness. □

This is the factor theorem for linear maps. It covers deterministic linear retentions only. For a stochastic sample retention the parent channel must be a Markov kernel, and its existence is the garbling condition stated with the channel square in the main text; a linear solution with negative entries does not qualify.

For example, let $C_j(u, v) = u$ and C_i be the identity on $\mathbb{R}$. The map $M_{ij}(u, v) = u + v$ cannot descend: the discarded vector $(0, 1)$ changes the retained output. The map $M_{ij}(u, v) = 2u$ descends to multiplication by two. Agreement of two current input laws does not change either conclusion, since the requirement concerns the boundary mechanism on its declared domain.

The same criterion handles an additive-noise channel without requiring Gaussian noise. Suppose the receiver content is $M_{ij}v + \varepsilon$, where ε has a fixed probability law ν independent of v. If eq. (S4.2) holds, its retained law is the distribution of

$$M^c_{IJ} C_j v + C_i \varepsilon. \tag{S4.3}$$

Thus the parent inherits the pushed noise law $(C_i)_{\#}\nu$. A state-dependent noise law requires its own factorization through C_j; factoring the mean map alone is insufficient.

S4.2 Exact finite filtering

Let X_n be a Markov chain on a finite configuration space S with transition matrix T and initial row distribution p_0. Let $Z_n = R(X_n)$ for a map into a finite retained space S^c. For each z, let P_z be the diagonal matrix whose entry at x is $1_{\{R(x)=z\}}$. These matrices are selectors of configurations, not new dynamical transitions.

Proposition S4.2 (The terminal conditional law retains every future prediction). For a retained history $h_n = (z_0, \dots, z_n)$ of positive probability, define

$$f_{h_n} = p_0 P_{z_0} T P_{z_1} \cdots T P_{z_n}, \qquad \widehat{p}_{h_n} = \frac{f_{h_n}}{f_{h_n}\mathbf{1}}. \tag{S4.4}$$

Then $\widehat{p}_{h_n}$ is the conditional distribution of X_n given h_n, and

$$\Pr(Z_{n+1} = z \mid h_n) = \widehat{p}_{h_n} T P_z \mathbf{1}. \tag{S4.5}$$

After a positive-probability next value z, the update is

$$\widehat{p}_{(h_n,z)} = \frac{\widehat{p}_{h_n} T P_z}{\widehat{p}_{h_n} T P_z \mathbf{1}}. \tag{S4.6}$$

For any finite future word, its probability is obtained by further multiplication by the corresponding TP_z matrices and by $\mathbf{1}$. The recursion is the forward algorithm for a function of a Markov chain [2, 5]. The normalized vector $\widehat{p}_{h_n}$ is itself a Markov process on the simplex of laws on S, moving to eq. (S4.6) with probability eq. (S4.5); it is the information state of a partially observed Markov chain [1]. The proof uses the Markov property at every propagation step, and time-dependent transition matrices T_n may replace T without change.

Proof. The x component of $p_0 P_{z_0}$ is the joint probability of $X_0 = x$ and $Z_0 = z_0$. Multiplication by T propagates this joint mass one step; multiplication by P_{z_1} restricts it to the next retained value. Induction gives the joint mass of h_n and $X_n = x$ as the x component of f_{h_n}. Its sum is the probability of h_n, so division gives the conditional distribution. One further propagation and selection gives eqs. (S4.5) and (S4.6). Repeating the operation gives every future word probability. □

A zero-probability history has zero unnormalized message and no conditional law supplied by this formula. Because all matrices are nonnegative, every extension of that history also has zero mass. The construction therefore needs no arbitrary prediction on an impossible history.

Remark S4.3 (The filter beyond finite configuration spaces). Let S be a measurable space, let the retained space Z be countable, let T_n be Markov kernels on S, and let R be measurable. For a retained history h of positive probability define the measures $f_{(z_0)}(B) = p_0(B \cap R^{-1}\{z_0\})$ and $f_{(h,z)}(B) = \int f_h(dx) T_n(x, B \cap R^{-1}\{z\})$. The argument of proposition S4.2 applies with measures in place of vectors: the normalized measure $f_h / f_h(\mathsf{S})$ is the conditional law of the current configuration

given h, the next retained value has probability $f_{(h,z)}(\mathsf{S})/f_h(\mathsf{S})$, and the update holds at every history of positive probability. When Z is uncountable, every history may have probability zero. On standard Borel spaces the conditional laws then exist as regular conditional distributions and the update is a jointly measurable disintegration of the one-step joint law of next state and next retained value [7]; the identities hold almost surely, choices of versions on null sets cannot be avoided, and the remark above on impossible histories has no analogue. The Markov property of the normalized filter holds once a jointly measurable version of the update is fixed. On an infinite configuration space the filter need not concentrate even when the retained history determines the initial configuration: for the doubling map $x \mapsto 2x \bmod 1$ with $R(x) = 1_{\{x \geq 1/2\}}$ and a uniform initial law, the conditional law of X_n given its retained history is uniform on an interval of length $1/2$ for every n.

The filter is sufficient even when the retained process is not Markov. For the deterministic cycle $1 \to 2 \to 3 \to 1$ with uniform initial law, set $R(1) = R(2) = 0$ and $R(3) = 1$. The histories $(0, 0)$ and $(1, 0)$ both have positive probability and end at the same retained value. The first ends at state 2 and predicts 1 next; the second ends at state 1 and predicts 0. Hence the current retained value cannot determine its next distribution. The terminal conditional law does: it is the point mass δ_2 after $(0, 0)$ and δ_1 after $(1, 0)$. Here the filter recovers the fine state, so it compresses nothing; a history-based parent forgets information only when noise or nonobservability prevents the history from determining the fine state.

S4.3 Successive retention of responses

Suppose a declared joint law contains a fine state X and a boundary response O. Let $Z = R_{10}(X)$ and $W = R_{21}(Z)$, and assume all spaces are standard Borel. For every bounded measurable response observable g, conditional expectation gives

$$\mathbb{E}[g(O) \mid W] = \mathbb{E}\big[\mathbb{E}[g(O) \mid Z] \mid W\big]. \tag{S4.7}$$

To verify the identity, integrate either side over any event determined by W. Such an event is also determined by Z, so both integrals equal the integral of $g(O)$ over that event. This characterizes conditional expectation up to null sets.

The identity concerns one declared joint law. Applying it to an equilibrium law produces equilibrium-conditional responses. Applying it to the joint law conditional on a retained history produces the corresponding predictive responses. These are interchangeable only when their relevant conditional laws agree. Exact boundary descent is stronger: it fixes the response on every admitted fiber, independently of the conditional weights used there.

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